Chapter 11: Rolling and Angular Momentum

The following subjects will be discussed

- Angular Momentum.
- Conservation of Angular Momentum.
- Examples

Angular Momentum

Angular momentum is the momentum associated with angular motion. It is defined with respect to some axis of rotation.

For a single particle:

$$\vec{l} = \vec{r} \times \vec{p}$$

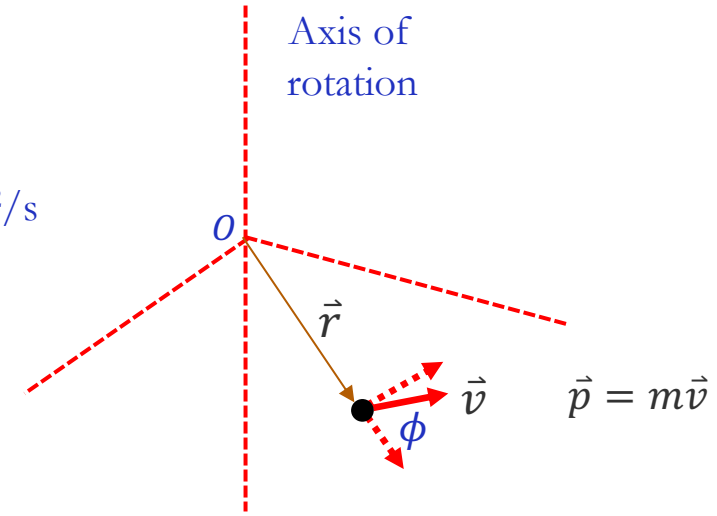
$$\vec{l} = m(\vec{r} \times \vec{v})$$

Kg m^2/s

Position vector relative to axial point

Linear momentum

$$|l| = m\,|\vec{r}||\vec{v}|\sin\phi$$



Angular momentum is in a direction normal to the plane of $\vec{r}$ and $\vec{v}$.

Counter-clockwise (CCW) rotation: Positive angular momentum.

Clockwise (CW) rotation: Negative angular momentum.

Note: Angular momentum is defined even with no rotation!!

$$\frac{d\vec{p}}{dt} = \vec{F}_{net}$$

$$\frac{d\vec{l}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \vec{F}_{net}$$

$$\frac{d\vec{l}}{dt} = \vec{\tau}_{net}$$

Angular Momentum

For Many particles:

$$\vec{L} = \vec{l}_1 + \vec{l}_2 + \cdots + \vec{l}_n = \sum_{i=1}^{n} \vec{l}_i$$

$$\frac{d\vec{L}}{dt} = \sum_{i=1}^{n} \frac{d\vec{l}_i}{dt} = \sum_{i=1}^{n} \vec{\tau}_i = \vec{\tau}_{net}$$

Over the same axis of rotation

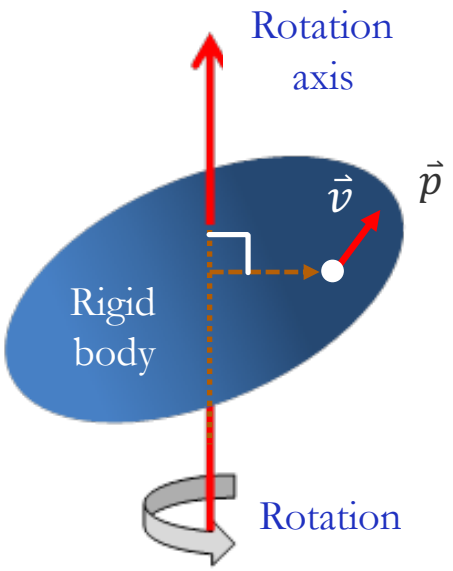
For a Rigid Body:

$$\vec{L} = \sum_{i=1}^{n} \vec{l}_i = \sum_{i=1}^{n} m_i \, (\vec{r}_i \times \vec{v}_i) = \sum_{i=1}^{n} m_i \, (\vec{r}_i \times \omega \vec{r}_i)$$

$$\vec{L} = \omega \sum_{i=1}^{n} m_i \, r_i^2 \qquad \vec{L} = I \, \omega$$

Over the same axis of rotation

$$\frac{d\vec{L}}{dt} = \vec{\tau}_{net}$$



Conservation of Angular Momentum

For a system with no net external torque ($\vec{\tau}_{net} = 0$), we have:

$$\frac{d\vec{L}}{dt} = 0 \quad \Rightarrow \quad \vec{L} = Constant$$

$$\vec{L}_{initial} = \vec{L}_{final}$$

Note that if $\tau_{net_x} = 0 \rightarrow L_x$ is conserved, and so on.

Interesting result

$$I_i\, \omega_i = I_f\, \omega_f$$

Change in shape (inertia)
causes
change in rotation speed.

Examples

1) A force $\vec{F} = (-8\vec{\imath} + 6\vec{\jmath})$ N acting on an object at a point located at $\vec{r} = (3\vec{\imath} + 4\vec{\jmath})$ m. Find the torque created around the origin.

$$\vec{\tau} = \vec{r} \times \vec{F} = (3\vec{\imath} + 4\vec{\jmath}) \times (-8\vec{\imath} + 6\vec{\jmath}) = 50\vec{k} \text{ N.m}$$

2) A 3-kg object is in motion such that its position at any given time is expressed as $\vec{r}(t) = 4t^2\,\vec{\imath} - (2t + 6t^2)\vec{\jmath}$ m. Find the torque around the origin at any time t.

$$\vec{r}(t) = 4t^2\,\vec{\imath} - (2t + 6t^2)\vec{\jmath}$$

$$\vec{v}(t) = 8t\,\vec{\imath} - (2 + 12t)\vec{\jmath}$$

$$\vec{l} = m(\vec{r} \times \vec{v}) = (3)\left[\left(4t^2\,\vec{\imath} - (2t + 6t^2)\vec{\jmath}\right) \times \left(8t\,\vec{\imath} - (2 + 12t)\vec{\jmath}\right)\right]$$

$$= 24t^2\,\vec{k}$$

$$\vec{\tau}_{net} = \frac{d\vec{l}}{dt} = 48t\,\vec{k} \text{ (N.m)}$$

Examples

3) The rotation period of the given structure is 2.5 second. Find the inertia of the structure and its angular momentum.

$$I = I_{loop} + I_{rods}$$

$$I_{loop} = \frac{1}{2}mR^2 + mR^2 \quad = \frac{3}{4} \text{ kg m}^2$$

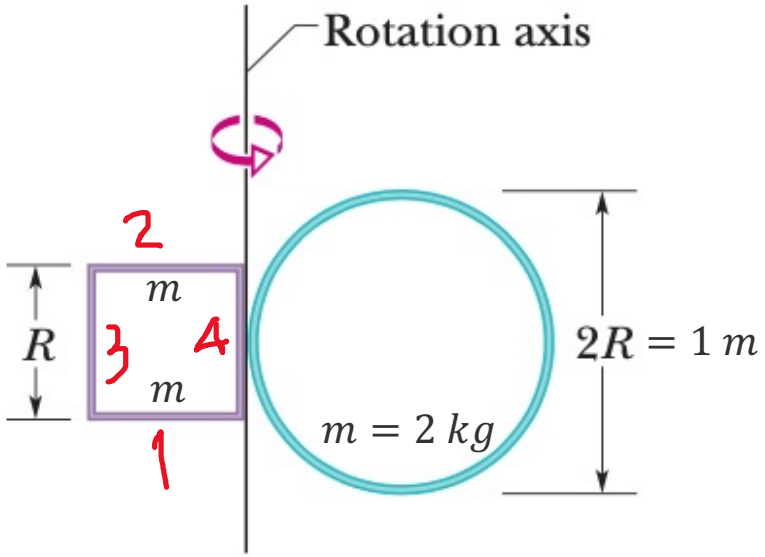
$$I_{rods_1_2} = \frac{1}{12}mR^2 + m\left(\frac{R}{2}\right)^2 = \frac{1}{3}mR^2$$

$$= \frac{1}{3} \text{ kg m}^2$$

$$I_{rods_3} = mR^2 \quad = \frac{1}{2} \text{ kg m}^2$$

$$I_{rods_4} = 0$$

$$I = 1.6 \text{ kg m}^2$$



$$l = I\,\omega = I\frac{2\pi}{T}$$

$$= (1.6)\left(\frac{2\pi}{2.5}\right) = 4.02 \text{ kg m}^2/\text{s}$$

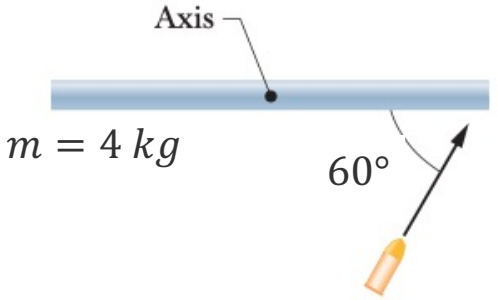
Direction?

Examples

4) The hanging rod is of length 0.5 meters and is initially at rest. A 3-g bullet with unknown speed hits and sticks into the rod. The angular speed of the rod after collision is 10 rad/s. Find the bullet's speed just before collision.

$$\vec{L}_{initial} = \vec{L}_{final}$$

Top view



$$\vec{L}_{initial} = l_b + l_r = l_b \qquad \vec{L}_{final} = l_{b+r} = I_{b+r}\,\omega_f$$

$$L_{initial} = l_b = m_b(\vec{r} \times \vec{v}) = m_b \left(\frac{L}{2}\right) v_b \sin 60$$

$$L_{final} = \left[\frac{1}{12} m_r L^2 + m_b \left(\frac{L}{2}\right)^2\right] \omega_f$$

$$v_b = \frac{\left[\frac{1}{12} m_r L^2 + m_b \left(\frac{L}{2}\right)^2\right] \omega_f}{m_b \left(\frac{L}{2}\right) \sin 60}$$

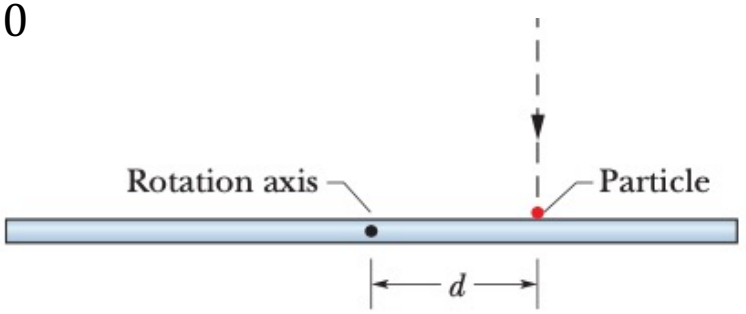
$$v_b = 1293.2\ m/s$$

Examples

5) The rod (length 0.6 m, mass m) is initially rotating CCW in a vertical axis with angular speed of 80 rad/s. A piece of gum (mass $m/3$) moves vertically downward and hits and sticks to the rod the moment it was horizontal at a distance d from the axis with speed at collision of 40 m/s.

After the collision, the rod stops rotating. Find d.

Side view



$$\vec{L}_{initial} = \vec{L}_{final} \qquad \vec{L}_{initial} = l_g + l_r \qquad \vec{L}_{final} = 0$$

$$L_{initial} = -\frac{m}{3}\, d\, v_g \sin 90 + \frac{1}{12} m L^2 \omega_r$$

Solve for d

$$d = \frac{\frac{1}{12} m L^2 \omega_r}{\frac{m}{3}\, v_g \sin 90}$$

$$d = 0.18\, m$$

Suppose the piece of gum hits at $d > 0.18\, m$, what would happen after collision?

Examples

6) An old plastic record (mass 0.1 kg, radius 10 cm) of inertia 0.5 g/m^2 is playing freely at an angular speed of 4.7 rad/s. A piece of gum (mass 20 g) is stuck to it at the edge. What happens immediately after that?

$$I_i\,\omega_i = I_f\,\omega_f$$

$$\omega_f = \frac{I_i}{I_f}\,\omega_i$$

$$\omega_f = \frac{I_i}{I_i + m_g R^2}\,\omega_i$$

$$= \frac{5\times10^{-4}}{5\times10^{-4} + (0.02)(0.1)^2}(4.7) = 3.36\,rad/s$$



Q & A

Send me an email if
you have any
questions.